higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N6

4 APRIL 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1
- Branch Currents.
- Branch Voltages. (2)

1.2 1.2.1

$$T = RC = 200 \times 10^{-3} \times 20 \times 10^{-6} = 4 \ s$$

snswer is option A ✓✓ (2)

1.2.2

$$V_C \ is \ \frac{85}{100} \times 200 = 170 \ V$$

Answer is option C ✓✓ (2)

1.2.3

$$V_C = E_M \left(1 - e^{\frac{-t}{T}}\right)$$

$$\frac{V_C}{E_M} = 1 - e^{\frac{-t}{T}}$$

$$\frac{V_C}{E_M} - 1 = - e^{\frac{-t}{T}}$$

$$\frac{170}{200} - 1 = - e^{\frac{-t}{T}}$$

$$-0.15 = - e^{\frac{-t}{T}}$$

$$0.15 = e^{\frac{-t}{T}}$$

$$\ln \ 0.15 \times T = \ -t$$

$$\ln \ 0.15 \times 4 = \ -t$$

$$-t = 7.5895 \ S$$

Answer is option B ✓✓✓ (3)

1.2.4

$$\omega = 0.5 C . V^2$$

$$= \ 0.5 \times 20 \times 10^{-6} \times 200$$

$$= 2 \ mJ$$

Answer is option D (2)

[11]

QUESTION 2

2.1 B
2.2 D
2.3 E
2.4 C
2.5 A

(5 × 2) **[10]**

QUESTION 3

3.1 3.1.1 False
3.1.2 False
3.1.3 False
3.1.4 True
3.1.5 False

(5 × 1) (5)

3.2 To allow the magnetostrictive core to expand✓ along the nodal line✓ so that the core can oscillate when an AC signal from the ultrasonic generator is applied✓ (3)

3.3
- Direct piezo-electric effect
- Converse piezo-electric effect

(2)
[10]

QUESTION 4

4.1

$$\% \ efficiency = 1.4 \times 2 \times V \times 100$$

$$\therefore V_{SUPPLY} = \frac{\% \ efficiency}{1.4 \times 10^{-9} \times 2 \times 100} \quad ✓$$

$$¿ \frac{6.468 \times 10^{-3}}{1.4 \times 10^{-9} \times 2 \times 100} \quad ✓$$

$$= 1.1 \text{ KV} \quad ✓$$

(3)

4.2

$$\lambda = \frac{h \times C}{E \times q}$$

$$¿ \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{1100 \times 1.602 \times 10^{-19}} \quad ✓✓$$

$$= -112.802 \times 10^{-11} m \quad ✓$$

(3)

4.3

$$C = \lambda \times f$$

$$f = \frac{C}{\lambda} \quad ✓$$

$$¿ \frac{3 \times 10^{8}}{-112.802 \times 10^{-11}} \quad ✓$$

$$= 266 \times 10^{15} Hz \quad ✓$$

(3)

$= 266\ PHz$

4.4

$$Total\ \ power\ \ dissipated = total\ \ power - power\ \ used$$
$$Total\ \ power = V \times I$$
$$but\ \ I = Cathode\ \ emission \times q$$
$$= 1 \times 10^{17} \times 1.602 \times 10^{-19}$$

16,02 µA ✓

$$\therefore Total\ \ power = 1100 \times 16.02\ \ \times 10^{-3}$$
$$= 17.622\ W \quad ✓$$

$$X\text{-}ray\ \ power\ \ used = Total\ \ power\ \ \times efficiency$$
$$= 17.622\ W\ \ \times 6.468 \times 10^{-3}$$
$$= 113.979\ mW \ ✓$$

$$\therefore Total\ \ power\ \ dissipated = total\ \ power - power\ \ used$$
$$= 17.622 - 113.979 \times 10^{-3} \quad ✓$$

$$= 17.508\ W \quad ✓$$ (5)

[14]

QUESTION 5

5.1 5.1.1 Rejected articles that can still be returned to the production line for reworking

5.1.2 Approved articles that are manufactured according to specifications

5.1.3 Rejected articles that have to be scraped, considered as waste products

(3 × 1) (3)

5.2 5.2.1 Indirect viewing method (1)

5.2.2 A X-ray tube
B Article being inspected
C X-ray film
D Screen (4)

5.2.3 Time-consuming due to the time needed to develop the film (1)

[9]

QUESTION 6

6.1
- Circuit analysis✓ – to determine the worst possible conditions✓
- Evaluation✓ – to ensure a margin of safety under the worst possible conditions✓
- Construction review✓ – to ensure that critical components are reliable, and that newly designed circuits function as planned✓ (3 × 2) (6)

6.2
- Transmitter
- Medium
- Optical receiver (3)

6.3 To prevent injury or death of workers OR to prevent damage to machinery. (1)
[10]

QUESTION 7

7.1
- High data transmission rate
- Universal language
- Communication interface
- Communication organisation (4)

7.2
- Overdamping✓ - Low gain✓
- Critical damping✓ - Low to medium gain✓
- Underdamping✓ - High gain✓ (6)

[10]

QUESTION 8

8.1 8.1.1

$$Average\ \ output\ \ voltage\ \ Vdc$$

$$but\ \ V_{dc}=\frac{V_{MAX}}{\pi}(1+\cos\alpha)$$

$$¿\ \ V_{MAX}=\sqrt{2}\,V_{RMS}$$

$$¿\sqrt{2}(230) \quad ✓$$
$$=325.269\ \ V \quad ✓$$

$$\therefore V_{dc}=\frac{V_{MAX}}{\pi}(1+\cos\alpha)$$

$$=\frac{325.269}{\pi}(1+\cos 30)✓$$
$$=193.201\ \ V✓$$ (4)

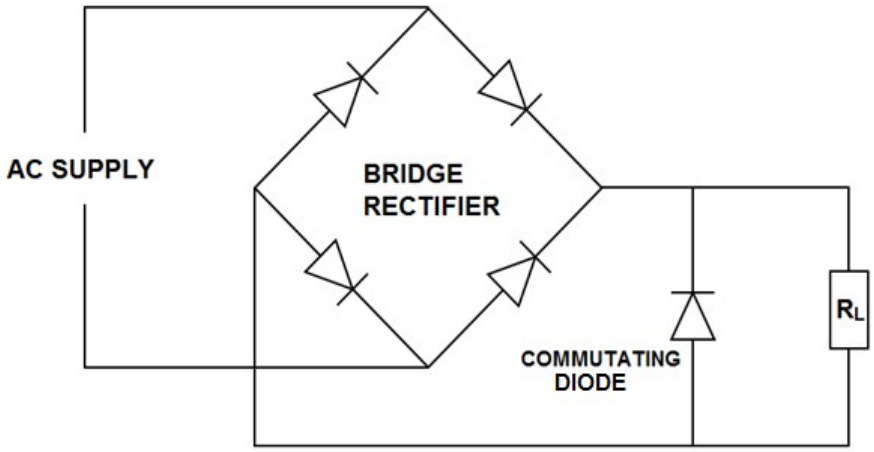
8.1.2

$$I_{DC}=\frac{V_{DC}}{R} \quad ✓$$

(3)

$$\therefore \frac{193.201}{10} \quad \checkmark$$
$$= 19.320 \ A \quad \checkmark$$

8.2



(4)

8.3 To transfer the load current away from the rectifier circuit✓ whenever the load voltage goes into a reverse state.✓ (2)
[13]

QUESTION 9

9.1
1 = Normally Open Contact
2 = Normally Closed Contact
3 = Direct Coil
8 = Inverted Coil
9 = End Statement (5)

9.2 Two rungs (1)

9.3 Group 1 combinations:
- Input 1 must be ON
- Input 2 must be OFF
- Input 4 must be OFF

Group 2 combinations:
- Input 4 must be ON (4)

9.4
- Input 7 must be ON✓
- Either input 5 or input 6 must be ON✓✓ (3)

[13]

TOTAL: 100